Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

In logistic regression, the predicted probability is

$$z(x) = \sigma\left(w^\top x + b\right).$$

The classification rule is:

$$\begin{array}{ll} \text{predict } 1 & \text{if } z(x) > 0.5 \\ \text{predict } 0 & \text{if } z(x) \le 0.5 \end{array}$$

Which of the following statements about the decision boundary are correct?

A. The decision boundary is defined by $w^\top x + b = 0$.

B. The ~~decision boundary is defined by~~ $w^\top x + b = 1$.

C. If $w^\top x + b < 0$, the classifier predicts label 0.

D. If $w^\top x + b > 0$, the classifier predicts label 1.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Choice Type Award: 1 Penalty: 0.33 Machine Learning

For the logistic regression model

$$p(x) = \sigma(4 - x)$$

consider the vertical decision boundary at $x = 4$.

What happens to the predicted probability $p(x)$ as we move leftward from $x = 4$ along the x-axis?

A. The probability decreases below 0.5 , so the model becomes more confident in class 0.

B. The probability increases above 0.5 , so the model becomes more confident in class 1.

C. The probability remains constant at 0.5 for all values left of 4.

D. The probability oscillates between 0 and 1 as x decreases.

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 04sec Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

In logistic regression with d features and no regularization ($\lambda = 0$), the loss for a single training example (x, y), where $y \in \{0, 1\}$, is:

$$\ell(w; x, y) = -\left[y \ln \sigma\left(w^\top x\right) + (1 - y) \ln\left(1 - \sigma\left(w^\top x\right)\right)\right]$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$.

Which of the following is the correct stochastic gradient update rule?

A. $w \leftarrow w + \eta\left(y - \sigma\left(w^\top x\right)\right) x$

B. $w \leftarrow w - \eta\left(y - \sigma\left(w^\top x\right)\right) x$

C. $w \leftarrow w + \eta\left(y + \sigma\left(w^\top x\right)\right) x$

D. $w \leftarrow w - \eta\left(y + \sigma\left(w^\top x\right)\right) x$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

In logistic regression with d features and no regularization ($\lambda = 0$), the loss for a single training example (x, y), where $y \in \{0, 1\}$, is:

$$\ell(w; x, y) = -\left[y \ln \sigma\left(w^\top x\right) + (1 - y) \ln\left(1 - \sigma\left(w^\top x\right)\right)\right]$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$.

What is the average time complexity of performing one stochastic gradient update using the correct rule in d-dimensional data?

A. $O\left(d^2\right)$

B. $O(\log d)$

C. $O(d)$

D. $O(1)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Consider a logistic regression model with weights $\beta = (\beta_1, \beta_2, \beta_3)$.
A given document has a feature vector $x = (1, -2, -1)$.

The logistic model predicts:

$$p(y = 1 \mid x) = \frac{1}{1 + \exp(-\beta^\top x)}.$$

Which of the following values of β will make the probability $p(y = 1 \mid x)$ close to 1?

A. $(1000, 0, 0)$

B. $(-1000, 0, 0)$

C. $(0, 1000, 0)$

D. $(0, -1000, 0)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

A researcher fits the following logistic regression model to study the probability that a student attends an optional review session (S) :

$$\Pr(S=1 \mid P)=\frac{1}{1+e^{-(\alpha+\beta P)}}$$

where P is the student's GPA in prerequisite courses.
The estimated coefficient for GPA is $\beta>0$.
Which of the following is the most appropriate interpretation of this result?

A. Students with higher GPA are less likely to attend the review session.

B. Students with higher GPA are more likely to attend the review session.

C. GPA has no effect on the probability of attending the review session.

D. Logistic regression coefficients cannot be interpreted in terms of likelihood.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 01sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Consider binary classification with the cross-entropy loss.
We are interested in the gradient of the loss with respect to the prediction $\hat{y}$:

$$\frac{\partial \operatorname{Loss}(\hat{y}, y)}{\partial \hat{y}}$$

Suppose the model predicts $\hat{y}=0.5$ and the true label is $y=1$.
What is the value of the gradient?

Your Answer: | Correct Answer: -2 | Not Attempted | Time taken: 00min 26sec | Discuss

Q #9 Multiple Select Type Award: 1 Penalty: 0 Machine Learning

We want to classify points using logistic regression. Consider the following two models:

- **Linear logistic regression:**

$$P(y = 1 \mid x, w) = \sigma(w_1 x + w_0)$$

- **Quadratic logistic regression:**

$$P(y = 1 \mid x, w) = \sigma\left(w_1 x + w_2 x^2 + w_0\right)$$

where $\sigma(\cdot)$ is the sigmoid function.

Suppose we are given three points with labels ± 1. Which of the following sets of points **cannot** be separated by the linear logistic regression model but **can** be separated using the quadratic feature model?

A. $(x = 1, y = 1),\ (x = 2, y = -1),\ (x = 3, y = 1)$
B. $(x = 1, y = -1),\ (x = 2, y = -1),\ (x = 3, y = 1)$
C. $(x = 1, y = 1),\ (x = 2, y = 1),\ (x = 3, y = -1)$
D. $(x = 1, y = -1),\ (x = 2, y = 1),\ (x = 3, y = -1)$

Your Answer: Correct Answer: A;D Not Attempted Time taken: 00min 02sec Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Which of the following statements about logistic regression is NOT true?

A. ℓ_2 (i.e., L^2) regularization can be added to the logistic regression cost function to prevent overfitting by penalizing large coefficient values.

B. The maximum likelihood estimates for the logistic regression coefficients can be found in closed form.

C. The logistic sigmoid function is used to model the probability of the positive class in binary logistic regression.

D. None of the above.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type Award: 2 Penalty: 0.67 Machine Learning

A logistic regression model for predicting survival on the Titanic has parameter vector

$$\vec{\beta} = [-1.2, \quad -0.005, 2.5]^T$$

where:

- $-1.2 =$ intercept,
- $-0.005 =$ coefficient for age,
- $2.5 =$ coefficient for sex (female $= 1$, male $= 0$).

Consider a passenger who is $\mathbf{20}$ years old and female. Suppose the model predicts survival probability using the logistic function

$$\hat{y} = \sigma\left(\vec{x}^T \vec{\beta}\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

and the passenger actually survived ($y = 1$).
What is the cross-entropy loss for this passenger?

A. $-\log(\sigma(1.2))$

B. $-\log(1 - \sigma(1.2))$

C. $\sigma(1.2)$

D. $\log(\sigma(1.2))$

Q #12 **Multiple Choice Type** **Award: 2** **Penalty: 0.67** **Machine Learning**

A logistic regression model for predicting survival on the Titanic is given in terms of log-odds:

$$\log \frac{P(Y=1 \mid \vec{x})}{P(Y=0 \mid \vec{x})} = -1.2 - 0.005(\text{age}) + 2.5(\text{sex}),$$

where sex = 1 for female and 0 for male.

Let m be the odds of survival for a male passenger of a given age, and let f be the odds of survival for a female passenger of the same age.

Which of the following relationships holds between f and m?

A. $f = m \cdot 2.5$

B. $f = m + 2.5$

C. $f = m \cdot e^{2.5}$

D. $f = m^{2.5}$

Your Answer: **Correct Answer: C** **Not Attempted** **Time taken: 00min 00sec** **Discuss**

Q #15 **Multiple Select Type** **Award: 2** **Penalty: 0** **Machine Learning**

Which of the following statements are true about logistic regression? Recall that the sigmoid function is defined as $\sigma(x) = \frac{1}{1+e^{-x}}$ for $x \in \mathbb{R}$

A. L2 regularization is often used to reduce overfitting in logistic regression by adding a penalty for large coefficient values

B. The logistic sigmoid function is used to model the probability of the positive class in binary logistic regression

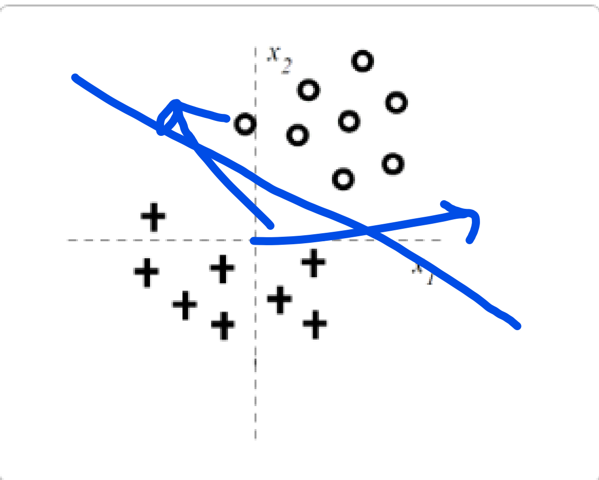
C. The maximum likelihood estimates for the logistic regression coefficients can be found in closed form.

D. For any finite input $x \in \mathbb{R}, \sigma(x)$ is strictly greater than 0 and strictly less than 1. Thus, a binary logistic regression model with finite input and weights can never output a probability of exactly 0 or 1 and can never achieve a training loss of exactly 0.

Your Answer: **Correct Answer: A;B;D** **Not Attempted** **Time taken: 00min 00sec** **Discuss**

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

We consider the binary classification task shown in the figure. The data is linearly separable, with class $y = 1$ (shown as '+') and class $y = 0$ (shown as 'O').



$$P(y = 1 \mid \mathbf{x}, \mathbf{w}) = \frac{1}{1 + \exp\big(-w_0 - w_1 x_1 - w_2 x_2\big)}.$$

We train *regularized* linear logistic regression models where we try to maximize

$$\sum_{i=1}^{n} \log\Big(P(y_i \mid x_i, w_0, w_1, w_2)\Big) - C w_j^2,$$

with the penalty applied to **exactly one parameter at a time**, i.e., $j \in \{0, 1, 2\}$.

For very large C, how does the training error behave when regularizing each parameter?

A. Regularizing $w_2 \longrightarrow$ training error increases.

B. Regularizing $w_1 \longrightarrow$ training error remains zero.

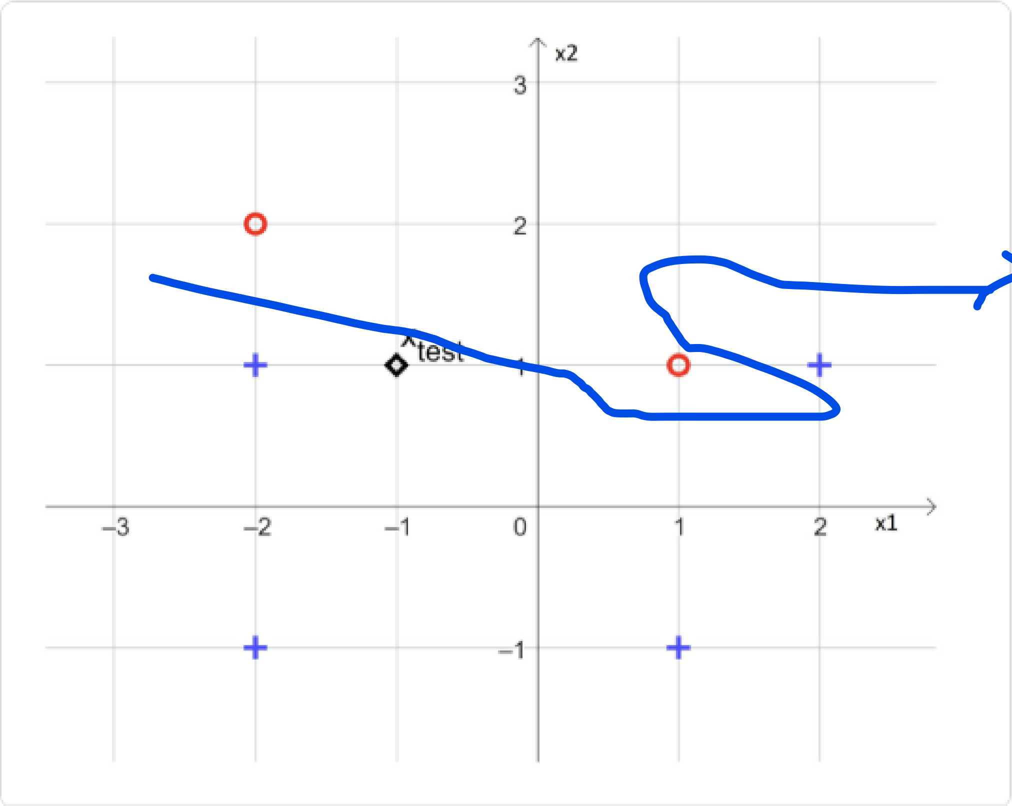
C. Regularizing $w_0 \longrightarrow$ training error increases.

D. None of the above.

Your Answer: | **Correct Answer: A;B;C** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 **Multiple Choice Type** **Award: 1** **Penalty: 0.33** **Machine Learning**

Emily is training a logistic regression classifier (with a bias term) on the 2-D dataset shown below.



Pluses (+) denote positives ($y = 1$) Circles (O) denote negatives ($y = -1$).

Logistic regression predicts $p(y = 1 \mid x) = \sigma(w^\top x + b)$ and its decision boundary is the line $w^\top x + b = 0$. Which statement best describes the boundary the model is likely to learn on this data?

A. The decision boundary will be a *nonlinear curve* that perfectly separates all points (zero error).

B. The decision boundary will be a *straight line*, and zero training error is achievable.

C. The decision boundary will be a *straight line*, but the data are not linearly separable, so at least one training point will be misclassified.

D. Logistic regression cannot be applied because the dataset has both positive and negative labels.